

Preliminary Results

Question 1: Which vehicle features affect resale price the most?

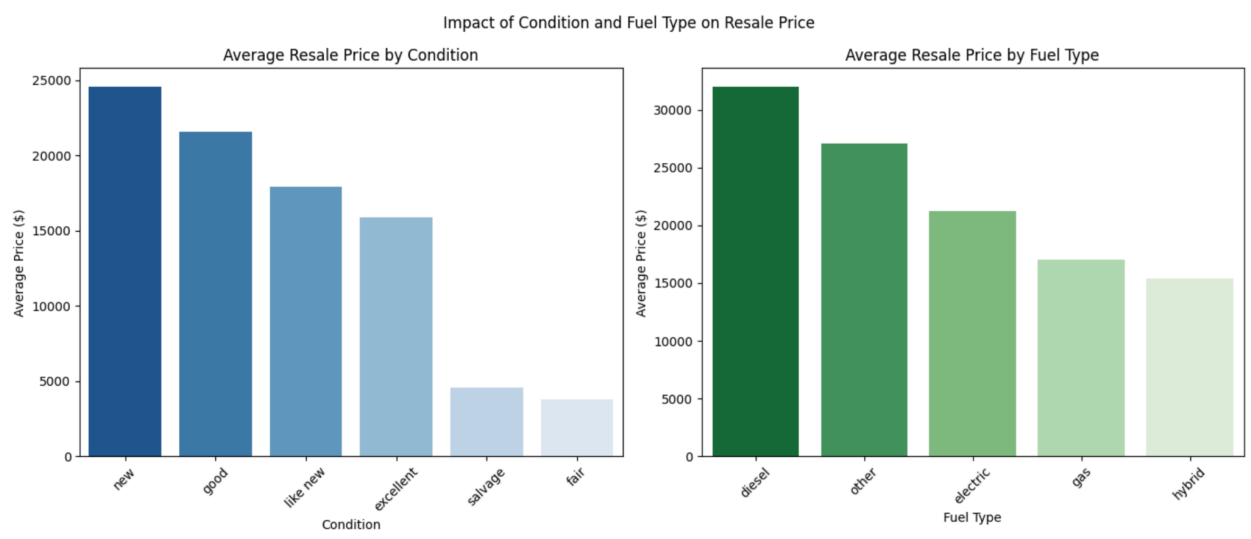


Figure 1. Impact of Condition and Fuel Type

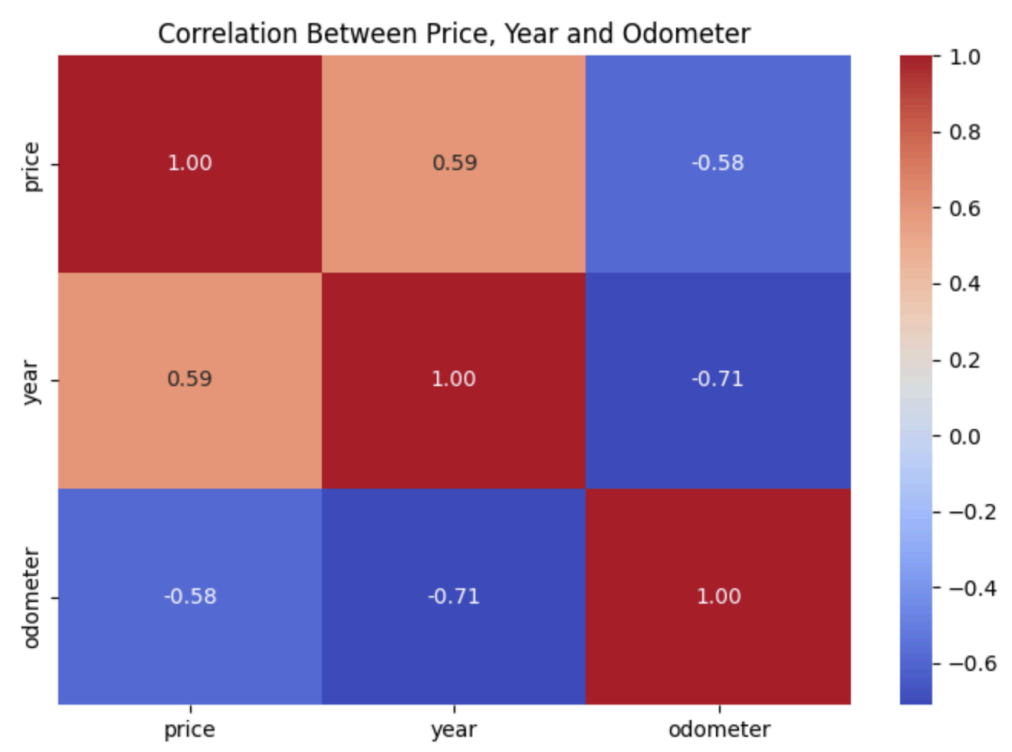


Figure 2. Correlation Between Price, Year, and Odometer

Summary

Our analysis explored how various vehicle features influence resale price in the California used car market. Based on correlation analysis, odometer reading and vehicle year showed the strongest relationships with price at -0.58 and 0.59 respectively, confirming that mileage and age are the primary drivers of resale value. The vehicle's condition also plays a significant role in resale price. New and good condition vehicles command the highest average prices at \$24,561 and \$21,573 respectively while fair and salvage condition vehicles drop dramatically to \$3,796 and \$4,576. This suggests that vehicle condition has a sharp impact on resale value especially at the lower end of the spectrum. Fuel type showed meaningful price differences across categories. Diesel vehicles averaged the highest resale price at \$32,001 which is consistent with our earlier finding that diesel trucks dominate the high end of the California used car market. Electric vehicles averaged \$21,195 while hybrid vehicles averaged the lowest at \$15,421 among conventional fuel types. Gasoline vehicles averaged \$17,050.

Geographically, Imperial County and Central Valley regions such as Hanford Corcoran and Bakersfield showed the highest average resale prices while urban areas like Los Angeles and San Diego showed lower averages. This is consistent with the truck heavy listings in agricultural regions driving prices up.

Implications

The condition findings suggest that vehicle condition plays a critical role in resale value. A vehicle in fair condition averages only \$3,796 compared to \$24,561 for a new condition listing which is a significant drop that buyers should keep in mind when purchasing a used car with the intent to resell later.

The diesel price premium is largely explained by the dominance of trucks in that fuel category rather than diesel passenger cars. This further supports our earlier finding that fuel type alone is not a reliable signal for price without also considering the type of vehicle.

Geographic price differences across regions appear to be influenced by the mix of vehicle types in each area rather than location alone. Our demand feature engineered from regional registration data will help us better isolate the true effect of geography on resale value in the modeling phase.

Future steps

- test model feature importance using XGBoost
- compare year and odometer with manufacturer, condition, fuel type, and region
- check whether these same patterns stay consistent across different vehicle segments
- use these results to improve the depreciation rating system proposed in the project

Question 2: How does geography in California affect resale value?

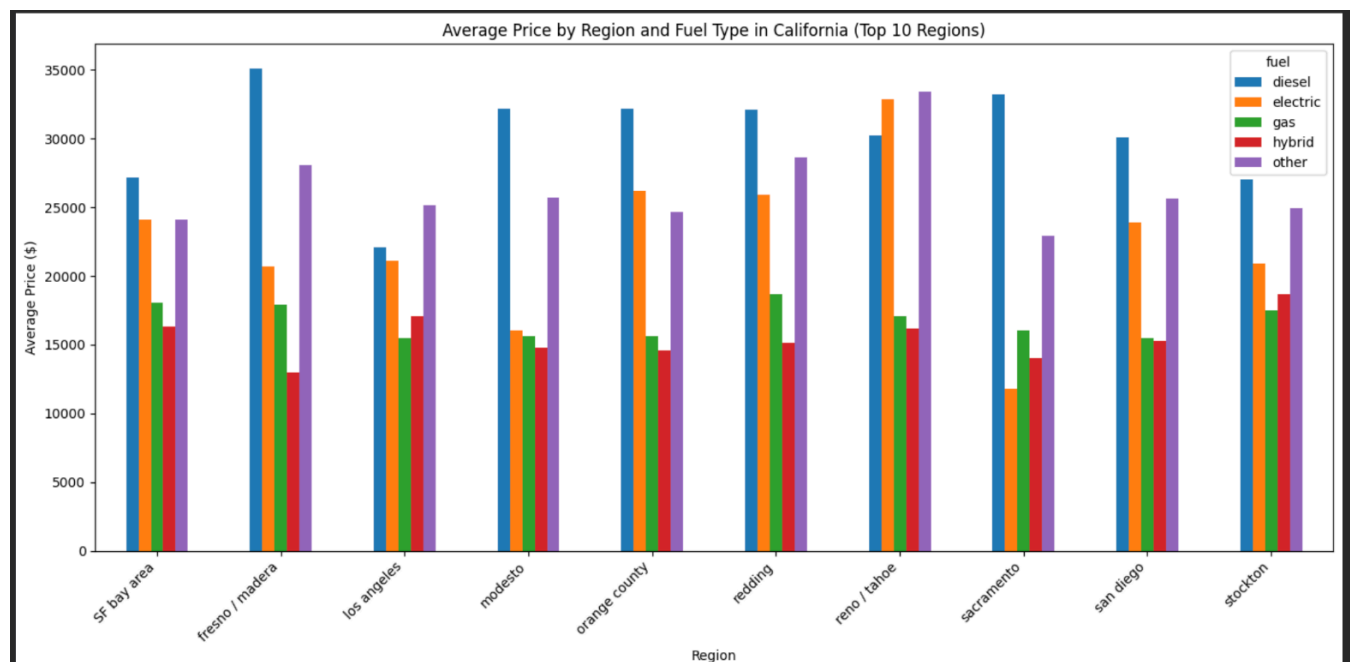


Figure 3. Average Used Car Price by Region and Fuel Type

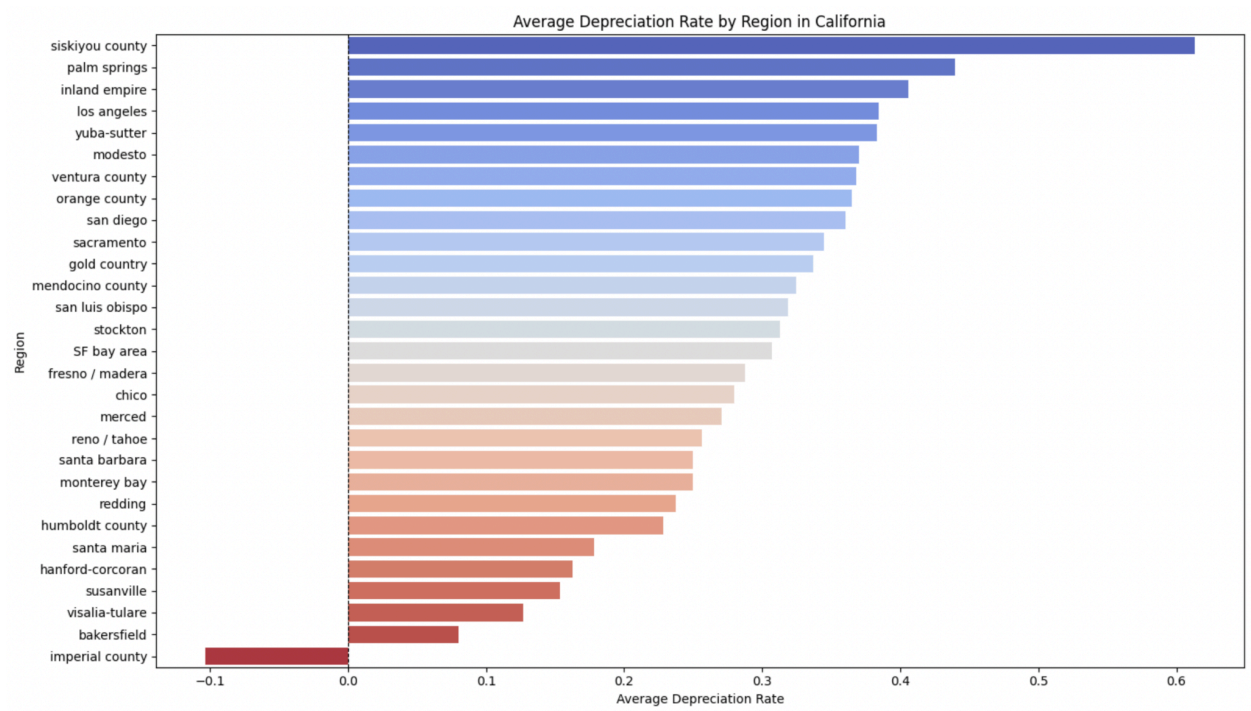


Figure 4. Average Depreciation Rate by Region in CA

Summary

Our analysis of depreciation rates across different California regions revealed some interesting geographic patterns in resale value. Siskiyou County showed the highest depreciation rate at 0.61 followed by Palm Springs at 0.44 and Inland Empire at 0.41. On the other end, Imperial County actually showed appreciation at -0.10 while Bakersfield and Visalia Tulare showed very low depreciation rates of 0.08 and 0.13 respectively.

Urban regions like Los Angeles and Inland Empire showed higher depreciation rates despite being wealthier areas. This is likely due to the higher volume of vehicle listings creating more competition in the used car market which drives prices down. In contrast Central Valley agricultural regions like Bakersfield and Imperial County showed much lower depreciation rates

which is consistent with our earlier finding that truck heavy listings in these areas tend to hold or appreciate in value.

The SF Bay Area sits in the middle at 0.31 which is interesting given its high income levels and strong EV adoption. This suggests that while EV demand is high in the Bay Area the overall used car market there is still competitive enough to keep depreciation rates moderate.

Implications

These findings suggest that geography does play a meaningful role in resale value in California though much of the geographic variation is influenced by the type of vehicles listed in each region rather than location alone. Agricultural regions have lower depreciation largely because of truck dominance while urban regions have higher depreciation due to higher supply and more competition.

Our demand feature engineered from regional registration data will help us better capture this geographic signal in our model. We expect that once we control for vehicle type and fuel type the geographic effect on depreciation will become clearer and more meaningful. As California continues shifting toward EVs we also expect to see increasing depreciation on gasoline vehicles in high EV adoption regions like the Bay Area in the future. This will be explored further in our regional clustering analysis in the next phase of the project.

Future Steps

- test whether region remains important after controlling for year, odometer, and manufacturer

- create region-specific depreciation curves for the website so users can compare local value retention more clearly
- Use K Means clustering to group regions with similar depreciation patterns for more accurate region specific estimates

Question 3: Are luxury, higher end brands depreciate faster than economical models such as a toyota or hybrid/electric cars? What kind of vehicles tend to maintain their resale value the most and do they fall into any specific category?

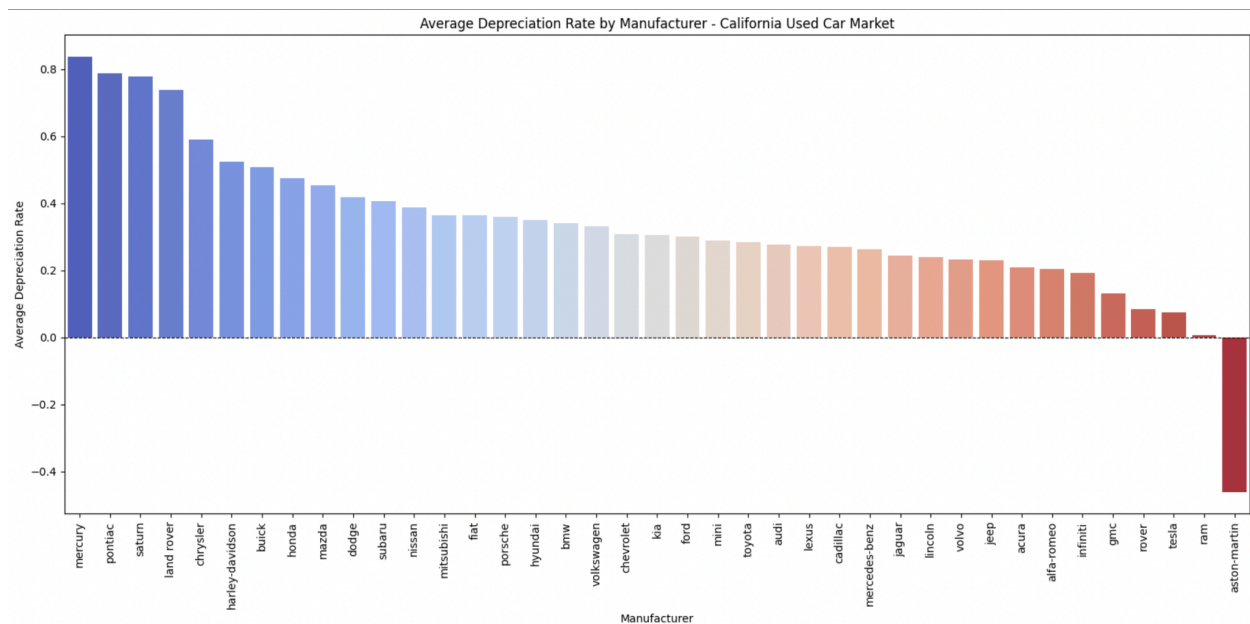


Figure 5. Average Depreciation Rate by Manufacturer, CA Used Car Market Data

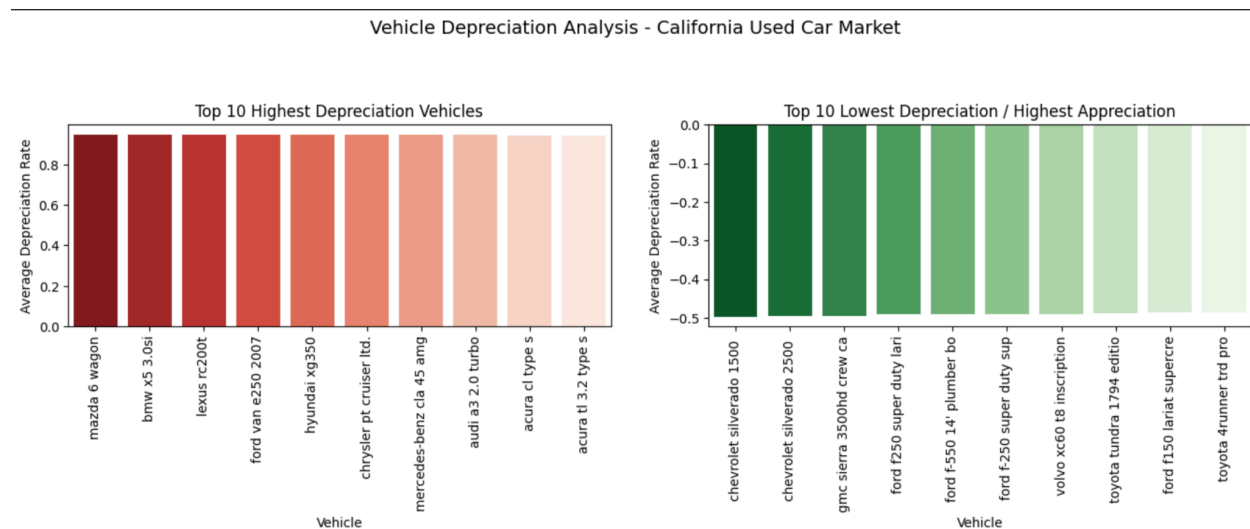


Figure 6. Top 10 Highest and Lowest Depreciation Chart For Different Make & Model

Summary

There are a few things that stood out to us after conducting some preliminary research. To our surprise, the depreciation rates of certain economical/everyday vehicles are slightly higher than some of the luxury or higher end models. This contradicts the common assumption that luxury vehicles lose value faster. Moreover, the vehicles with the highest appreciation and lowest depreciation rate in CA were trucks such as RAM, Ford F-150 and Silverado. For a passenger vehicle, Tesla held a significantly better depreciation rate of around 0.08, hinting at the growing demand for electric vehicles in CA. On the other hand, discontinued brands such as Pontiac, Mercury and Saturn depreciated significantly higher, suggesting that buyers have low demand for vehicles with potentially limited manufacturer support and parts availability.

Implications

The low depreciation rate of trucks in California can be a result of several factors. California has massive agricultural, construction and logistics industries such as the Central Valley farming and new housing projects which sustain consistent demand for work vehicles. Post 2020 supply chain disruptions also limited new truck inventory which drove used truck prices above their original MSRP in many cases.

The higher depreciation rate of certain economical vehicles can be a result of using aggregated vehicle MSRP prices instead of the true MSRP values for the associated model. This could have potentially inflated the MSRP prices and resulted in higher depreciation values. To remedy these artificial depreciation values, we are considering using an average MSRP price specific to each model instead of using the average price for the make as a whole to get a more accurate picture. Additionally as California continues shifting toward EVs, gasoline vehicles in high EV adoption regions may face increasing depreciation pressure over time which will be explored further in our regional analysis.

Future Steps

- Implement model level MSRP averaging to fix the inflation issue affecting economy brands like Honda and Nissan
- Separate trucks from passenger cars in the depreciation analysis since trucks are significantly skewing results across all brands